8 MITIGATION MEASURES

Iluka has a general environmental duty (GED) under the *Environmental Protection Act 2017* (EP Act) to eliminate or minimise the risk of harm from construction or operation activities to human health and the environment and from pollution and waste. This includes biodiversity and ecological values.

The project will be undertaken in accordance with an approved work plan issued under the *Mineral Resources (Sustainable Development) Act 1990* (MRSD Act) and in accordance with the environmental management framework (EMF) for the project. Both the work plan and EMF will require the detailed design of the project to avoid or minimise impacts to native vegetation and habitat to the extent practicable within the Disturbance Footprint.

The main biodiversity impacts expected from the project will occur because of land clearance during establishment of the mine layout as well as construction of ancillary infrastructure (ie. utility infrastructure including powerline and water pipeline).

The focus of mitigation measures is primarily to avoid impacts and secondly to develop and implement project-specific measures that will reduce impacts to acceptable levels. Thirdly, the project will progressively rehabilitate the land in consultation to establish a post-mining land use capable of supporting a productive agriculture commensurate with surrounding areas.

The key mitigation and management measures that will be addressed to manage the project risks (Table 8) are discussed below. The measures outlined in this section are known to be effective or best practice for managing impacts on ecological values.

8.1 Mitigation Measures

8.1.1 Awareness of ecological values

8.1.1.1 Inductions (EM01)

All contractors/staff will be inducted to site. The induction will outline the presence and location of ecological values within the works area, and the relevant protective measures and obligations while undertaking construction and mining activities.

8.1.1.2 Native Vegetation Protection (EM02 - EM04)

The project will define zones where works are not permitted to protect areas of ecological value and avoid impacts from inadvertent damage through mechanisms such as earthworks, vehicle parking, equipment and plant storage and stockpiling of soils and construction materials. The project will establish two types of zones to protect retained ecological features:

- establish No Go Zones (NGZs) to protect retained native vegetation patches. No-Go zone fencing will be erected:
    - along the edge of the Disturbance Footprint when located within patches of native vegetation;

  - within two metres of native vegetation where the Disturbance Footprint is located within 15 metres proximity to patches of native vegetation;

- Tree Protection Zone (TPZ) fencing. TPZ fencing will be erected along the edge of the TPZ of a scattered tree where the Disturbance Footprint is located within 15 metres proximity to the tree

TPZs will be established in accordance with the A*ustralian Standard 4970-2009 Protection of trees on development sites* (Standards Australia 2009).

A register identifying the locations of NGZs and TPZs will be created and maintained, with NGZs and TPZs clearly marked on all plans for the project and be incorporated into the Biodiversity Management Plan (EM03), and regular inspections and maintenance of fencing throughout construction and operation will be undertaken to ensure continued integrity will be undertaken.

8.1.1.3 Marking of Access Tracks and Roads (EM23)

All access tracks and roads must be clearly marked to prevent the establishment of secondary tracks and indirect native vegetation impacts.

8.1.1.4 Post-clearing Auditing (EM22)

A post native vegetation clearing audit at the end of each stage of development will be undertaken to reconcile the extent of native vegetation impacted during operation and construction against the permitted losses.

8.1.2 Measures to minimise impacts on retained vegetation and habitat

The following mitigation measures will be implemented to minimise and mitigate effects from the construction and ongoing operation of project components on retained native vegetation and habitat in addition to the commitment to works being restricted to the construction footprint of each component and the designation of NGZs and TPZs (EM02; EM03 and EM04).

Iluka will ensure that best practice sedimentation and pollution control measures are undertaken at all times in accordance with Environment Protection Authority guidelines to prevent offsite impacts to waterways and wetlands.

8.1.2.1 Erosion and Sediment Control (EM24)

The project will implement appropriate sediment and erosion control measures prior to any ground-disturbance works and throughout construction and operation to protect retained native vegetation and waterways/waterbodies.

Constructed landforms and sedimentation basins must be designed to minimise erosion and sedimentation to off-target areas.

8.1.2.2 Dust Management (EM16; EM25; EM27)

Dust control measures will be implemented and applied by the project. Standard controls relating to air quality and dust management that will be applied to the project are:

- use of water sprays on unsealed roads and surfaces.

- dust suppression sprays and sprinklers.
- cessation of works during high wind conditions.
- implement speed limits on vehicular traffic along haulage and internal access roads/tracks.

8.1.2.3 Placement of stockpiles, machinery and roads (EM40)

Stockpiles, machinery and infrastructure have been designed to be located away from waterways, dams and areas supporting native vegetation as much as practical, to reduce risk of disturbance.

8.1.2.4 Fire Management (EM33)

A bushfire risk assessment to identify potential impacts and appropriate mitigation and management measures will be undertaken.

Operational controls for fire management, including emergency management responses will be prepared and implemented and included in the BMP.

8.1.2.5 Seed Collection (EM17)

Collect seed from specimens of national and State significant flora species to ensure genetic diversity is not lost. While the focus is on collecting seed from threatened flora, seed can be collected from non0threatend flora as well, with seed to be used to propagate specimens for rehabilitation and revegetation works. Any additional seed material to be lodged with the RBGV seedbank.

8.1.2.6 Chemical Spill Management (EM26)

Maintenance activities and refuelling will be carried out at an appropriate distance with appropriate spill protection measures in place to avoid impacts to vegetation, habitat, and waterways, in accordance with relevant regulatory requirements.

Regulations, guidelines and best practice measure must be adhered to for the storage, handling, transport and use of materials that may contaminate the environment including (but not limited to):

- *Environment Protection Act 2017*;
- Dangerous Good (Storage and Handling) Regulations 2012;
- Liquid storage and handling guidelines (publication 1698);
- The Storage and Handling of Flammable and Combustible Liquids – Australian Standard 1940:2017; and
- advice from WorkSafe Victoria

8.1.2.7 Mitigate impacts to Red Gum Swamp, Darragan Swamp and Jallumba Marsh Flora and Fauna Reserve (EM28 - EM32; EM42)

Although not directly impacted by the project, the ecological values within Red Gum Swamp and Darragan Swamp have the potential to be indirectly impacted due to alterations to the catchment that informs surface water flows into these areas. Although the alterations to the catchment that feeds into Jallumba Marsh Flora and Fauna Reserve can be mitigated through the relocation of stockpiles, it is located immediately adjacent to the Disturbance Footprint.

Measures to mitigate impacts to these waterbodies will include:

- regulate supplementary flows into Red Gum Swamp and Darragan Swamp to mimic natural variability using an alternate water source. Supplementary water flows to be in accordance with the Wetland Watering Plan (Water Technology 2025) and reflective of seasonal conditions within the broader region.
- water testing must be undertaken bi-annually in Red Gum Swamp and Darragan Swamp in the years after supplementary water has been supplied to ensure no detrimental impacts to water quality and fauna habitats
- annual monitoring of ongoing habitat use by wetland-dependent fauna (frogs, waterbirds, turtles) will be undertaken at Red Gum Swamp and Darragan Swamp.
- annual monitoring of vegetation condition and extent will be undertaken at Red Gum Swamp, Darragan Swamp and Jallumba Marsh Flora and Fauna Reserve.
- during rehabilitation of the mine, final landform and drainage flows will be designed and constructed to re-instate pre-disturbance drainage patterns and catchments.

8.1.2.8 Groundwater Monitoring and Management (EM34; EM35)

A Groundwater Monitoring Plan will be prepared which will include monthly groundwater level monitoring in Project Area to detect extent and risk of changes in groundwater levels and groundwater quality.

The Groundwater Monitoring Plan will also include appropriate mitigation and management measures in the event that a risk is identified. The Groundwater Monitoring Plan will form a part of the BMP.

8.1.3 Measures to prevent introduction of weeds and pathogens

The project will progressively establish mining cells over the life of the project. Land stripping and stockpiling of materials will form part of the proposed works which could introduce weeds and pathogens to the Project Area if not managed appropriately.

The objective of measures presented below is to reduce the risk of weeds or pathogens being introduced to the site. It is proposed that this practice will continue throughout all phases of the project. All soil should be assumed to contain weed propagules and be managed accordingly. All water and damp soil should be assumed to be infected with pathogens and managed accordingly, unless testing shows otherwise.

8.1.3.1 Hygiene Controls (EM01; EM08; EM41)

Measures to ensure opportunities for the introduction and spread of weeds and pathogens, and the importation of weeds (importation of seeds and other vegetative material to the site) are limited will include:

- prepare and implement a Biodiversity Management Plan (BMP), which will include measures to manage and control impacts on biodiversity from biosecurity threats (weeds, pathogens and pest animals). These measures will include, but not be limited to:

- a requirement for all staff and contractors to be inducted on the presence and location of significant ecological values and inform them of all relevant protective measures and obligations;
- establishment and maintenance of wash-down locations for machinery, vehicles, tools and footwear
- ensure that gravel, crushed rock or other non-soil substrate imported to site is free from noxious weed seed
- identify known areas of significant weed infestations (noxious weeds listed under the *Catchment and Land Protection Act 1994* (CaLP Act) on site plans and in contractor induction material.
- drive on surfaced or defined roads and tracks.
- avoid driving through highly infested areas of weeds, in water or areas of wet soil, where possible.
- all noxious weed infestations must be managed and prevent the spread of these species into adjacent land.

8.1.3.2 Weed Management (EM09)

Prepare a Weed Management Plan. Weed controls will include:

- treat high risk weeds prior to works commencing;
- engage experienced land management contractor/s familiar with native plant and weed identification and possess a detailed working knowledge of herbicide selection and use and mechanical removal techniques to undertake weed control.
- implement a monitoring and control program for noxious weed species.
- conduct inspections to check for evidence of weed invasion, degradation or erosion near work areas.
- manage any outbreak of CALP Act and/or Weeds or National Environmental Significance (WoNS) that occurs due to construction activity. Prevent spread into adjacent land.
- ensure induction of all contract staff details the requirements for vehicles and equipment to be free of mud and plant material prior to entering work sites.

8.1.4 Measures to minimise disturbance, injury or death to wildlife

Measures to reduce fauna mortality and displacement during construction and operation are outlined below. By implementing these measures, the risk of fauna mortality and displacement would be reduced, and risk of residual impact would be low.

8.1.4.1 Pre-clearance Surveys and Fauna Salvage (EM05 - EM07)

A pre-clearance survey protocol and wildlife salvage plan will be developed in consultation with DEECA, and included in the BMP. A qualified zoologist must be present to conduct pre-clearance searches in any identified fauna habitats.

Where fauna habitat is identified, a suitably qualified wildlife handler ('wildlife spotter') holding a relevant and current authorisation under the *Wildlife Act 1975*, to salvage and relocate any wildlife encountered during site clearance work).

An Injured Wildlife Protocol will be developed and implemented in consultation with DEECA and wildlife carers which will outline steps to take in the event of injured wildlife being encountered and will provide contact details for local wildlife carers.

8.1.4.2 Timing of Habitat Removal (EM14)

Habitat removal will be undertaken in a staged manner, where appropriate, to allow fauna to disperse away from the construction footprint.

Any removal of habitat trees or shrubs (particularly hollow-bearing trees) will be undertaken between February and September to avoid the breeding seasons for most fauna species, where practicable.

Habitat trees or shrubs will not be removed between October and January (inclusive) to avoid the breeding season of most nesting birds and mammals, unless in accordance with protocols detailed in the Biodiversity Management Plan.

8.1.4.3 Replacement Hollows (EM12)

Nest boxes will be installed in adjacent retained habitat prior to tree removal) to compensate for the removal of hollow-bearing trees. At a minimum, the number of nest boxes installed will exceed the number of hollow-bearing trees removed. Relocation and salvage of suitable habitat (i.e. logs, trunks, stags) that will to be impacted or removed within the Disturbance Footprint will be repurposed and relocated for use as habitat outside of the Disturbance Footprint.

The nest boxes, (artificial wood, 3-D printed nest boxes, or carved logs) should be fixed securely to the trunk or a sturdy branch of the tree at least two metres above the ground, but at a height where it can be monitored easily, and is out of the prevailing wet-weather winds, in the shade or semi-shade, and near sources of food and water. Any existing hollows present within trees that are proposed to be impacted within the development footprint should be re-installed within retained vegetation outside of the impact area.

Investigations into viable hollow replacements will be considered in the post approvals and will be detailed in the BMP (or appropriate management plan) with consideration of the selected method's site-specific effectiveness for the target species.

Artificial Nest Boxes

The use of traditional artificial nest boxes has often found to be under-utilised (McNabb and Greenwood 2011). Further, many nest boxes set up to offset previous habitat loss fail to attract native animals due to them being utilised by non-native fauna (Lindenmayer *et al.,* 2017), and they also tend to disintegrate and become unusable after only a few years. However, novel approaches to nest box design such as 3-D printed nest boxes and the installation of carved hollows have been trialled in recent years.

3-D Printed Nest Boxes

A recent development that has been trialled in the City of Knox has been the creation and installation of 3-D printed nest boxes – created specifically to mimic the naturally occurring characteristics of known Powerful Owl nesting sites. These nest boxes can be moulded to a unique fit from a range of organic materials including hemp concrete, wood earth (i.e. clay, mud) and/or fungus, and as they are lightweight, they can be easily fixed onto trees.

Installation of Carved Hollows and/or Logs

The creation of carved hollows into tree trunks has been shown to better mimic the physical and thermal properties of natural tree hollows compared to artificial nest boxes and log hollows (Terry *et al.*, 2021). Artificial hollows carved directly into live trees can produce thermally stable supplementary habitats that could potentially buffer hollow-dependent fauna from weather extremes, whereas poorly insulated plywood nest boxes produce lower-quality thermal environments.

Installation of carved hollows and/or logs provides an opportunity to create long-term, sustainable ecologically robust habitat that increases the chances of successful breeding outcomes for hollow-dependant fauna.

8.1.4.4 Fauna Escape Mechanisms (EM13)

Trenching works will manage any open pits or trenches to reduce potential for fauna entrapment in accordance with the *APGA Code of Environmental Practice: Onshore Pipelines* (AGPA 2022) through measures such as:

- minimising the length of trench open at a time.
- minimising the amount of time trenches and other excavations are open.
- constructing trench plugs (short section of trench left unexcavated to allow passage of stock or wildlife across the trench) with slopes less than 45 degrees to provide exit ramps for fauna. Provide other exit ramps where practicable and safe to do so. Exit ramps to be installed at the ends of cable trenches.
- creating 'ladders' to enable fauna to exit the excavations (e.g. branches, ropes, planks, floats).
- daily inspections of open trenches by appropriately certified personnel to remove trapped fauna as required.
- checking for fauna prior to backfilling trenches.
- ensuring fauna are discouraged from work areas by erecting barriers where practicable.
- considering installing signage along access routes through conservation reserves to raise awareness of presence of wildlife crossing roads and reduce incidence of wildlife collision.
- developing and implementing a procedure for finding trapped fauna.

8.1.4.5 Wildlife-friendly Fence Design (EM15)

The best way to minimise entanglement or wildlife damage to fencing is to ensure the animals can see the fence (DELWP 2019). Where possible, fencing will be permeable to allow wildlife to move through the landscape. The exception to this is where areas are hazardous to wildlife, in which case fencing will seek to prevent access (ie. chain mesh fencing) if practicable.

Permanent No Go Zone fencing will comprise post and wire fencing that does not include barbed wire, or chain mesh fencing at least 1.8 m high to deter kangaroos from attempting to jump the fence if required (DELWP 2019). A white wire strand will be added at the top of the fence to make it more visible. The application of these two designs (post and wire or chain mesh) will depend on site-specific requirements to be determined on a site-by-site bases prior to construction.

Short-term, more temporary NGZs or other fenced areas will comprise a single wire strand with bunting flags rather than orange mesh para-webbing to reduce risk of entanglement.

8.1.4.6 Fauna Sensitive Lighting Design (EM11)

Any externally visible artificial lighting will consider the potential effects on wildlife. In areas of potential disturbance to sensitive fauna species the following measures will be implemented in accordance with the *National Light Pollution Guidelines for Wildlife* (DCCEEW 2023a):

- use of lighting technology that limit the spread of light outside of the works area and into adjacent habitat including keeping lights close to the ground and shielding lights to avoid light spill beyond the works area or facility.
- using a light beam that is minimally attractive to flying insects including use of red lamps or filters. Facilities established within or close to sensitive habitats will incorporate use of low intensity lighting and longer wavelength lighting such as red/orange (when not conflicting with safety requirements). Low-pressure sodium lamps and amber LEDs are low in the blue and UV wavelength emissions that attract insects and are generally less disruptive for wildlife (DCCEEW 2023a).

8.1.4.7 Traffic Management (EM16)

Measures to reduce the risk of wildlife-vehicle interactions will be implemented, including:

- implement speed limits on vehicular traffic along haulage and internal access roads/tracks.
- maintain agreed speed limit on access roads to the site as per occupational health and safety requirements.
- install signage warning of the presence of wildlife and ensuring signage is maintained.
- record vehicle-wildlife interactions to inform review of management measures.
- installation of canopy bridges over access / haulage roads to facilitate movement of arboreal fauna between habitats.

8.1.4.8 Noise and Vibration Abatement (EM36 - EM39)

Although the mine is in close physical proximity to areas of ecological sensitivity (ie. Jallumba Marsh Flora and Fauna Reserve, Darragan Swamp, Red Gum Swamp), there are several mitigation measures that will be implemented to mitigate against noise and vibration impacts during the construction and operation of the project. These include:

- preparation of a Fauna Management Plan that identifies important areas of habitat for fauna (ie. breeding habitat for Growling Grass Frog) and associated timing in order to mitigate high impact noise and/or vibration activities at key times of year, where required.
- where operations or equipment that emit excessive noise or vibrations, control measures such as baffles or mufflers will be employed.
- limit high impact noise events to daylight hours, and outside of key ecological times of the year where practical.
- traffic noise levels will not exceed the objectives specified in the VicRoads Traffic Noise Reduction Policy.

8.1.4.9 Maintenance and Enhancement of Fauna Connectivity and Habitat (EM15; EM18 - EM21)

As vegetation and habitat within the Project Area will be progressively removed, there will therefore be an opportunity to progressively rehabilitate and revegetate areas disturbed by the project with locally indigenous species. A summary of measures to mitigate against fauna habitat connectivity being compromised are summarised below:

- establish fauna-friendly fencing to facilitate movement throughout the Project Area
- revegetation activities outside the mine and infrastructure footprint should commence as soon as practical to reduce the delay on habitat availability (post-clearance).
- prioritise the creation of habitat corridor adjacent to the mine footprint to facilitate east/west connectivity to compensate for loss of the existing habitat corridor along Jallumba-Mockinya Road (Figure 7b).
- retain timber (trunks and logs) from cleared native vegetation for re-use in retained areas and rehabilitated area for use as additional habitat
- retain rocks and relocate to adjacent retained areas to provide habitat for fauna species such as reptiles.

Whilst the mine layout plan is largely defined by the extent of the resource, there will be ongoing opportunities to further avoid impacts to native vegetation (including several scattered native trees) and fauna habitat at a local scale, as the alignment and final siting of project infrastructure are further refined.

A detailed assessment of individual Large Trees, scattered trees and the associated presence of fauna habitat was undertaken, with the aim of retaining ensuring these values could be retained where possible.

8.1.5 Management Plans

Implementation of measures associated with the minimisation of impacts relies on the development of management plans to detail the measures, timeframes and performance objectives and responsibilities.

As part of the ongoing project planning process, detailed contingency and mitigation measures must be developed and presented within the project Risk Management Plan.

The Risk Management Plan must incorporate a Biodiversity Management Plan (BMP) relating to the management and monitoring of biodiversity during the construction, operation and rehabilitation phases of the project. Within the BMP, several sub-management plans will be included, addressing specific ecological values and associated mitigations, monitoring and management measures.

Management Plans (including the Biodiversity Management Plan) will be provided to ERR, DEECA and DTP for approval, who will have an opportunity to review and comment of the contents of the Plans as required. The required management plans can then be updated as required as part of the approval process.

The Biodiversity Management Plan will set out accountabilities and systems to manage and monitoring environmental effects and the approach to evaluating and reporting environmental outcomes and performance, and include ongoing biodiversity monitoring program to be implemented prior to the commencement of construction and during operations. The monitoring program will focus on areas of retained ecological value and include the following:

- enable an assessment of the quality and extent of retained vegetation against baseline data (i.e. data in this report) so any changes in the condition of habitats and vegetation can be quantified;
- determine thresholds of change to trigger management responses and ensure the thresholds are sensitive enough to prevent irreversible damage; and,
- detail contingency measures to be implemented if the project results in adverse residual effects to retained flora and fauna values.

The management plans relevant to ecological values to be prepared include:

- Biodiversity Management Plan (this is the overarching ecological management plan for the project)
- Flora and Fauna Management Plan
- Weed and Pest Animal Management Plan
- Threatened Species Protection Management Plan
- Pre-clearance Protocol and Fauna Salvage Plan
- Wetland Watering Plan

- Rehabilitation and Revegetation Plan
- Groundwater Monitoring Plan
- Offset Management Plan (Section 9.2).

8.2 Summary of Mitigation Measures

Specific mitigation measures recommended for implementation to alleviate impacts to biodiversity that have been identified as part of the risk assessment (Section 5.1) are detailed below (Table 23). Each predicted risk has been assigned one or more mitigation measures to reduce the risk associated with each potential impact.

Table 23. Proposed mitigation measures.

Measure	Mitigation ID	Action
Environmental Induction	EM01	Induct all employees and contract staff on the presence and location of significant ecological values and inform them of all relevant protective measures and obligations. Maps identifying all areas of ecological sensitivity must be provided during the induction.
No Go Zones	EM02	Establish zones to protect retained areas of native vegetation, threatened ecological communities, threatened species, and/or areas identified as important for connectivity: - Establish No Go Zones (NGZs) to protect retained native vegetation patches. - Tree Protection Zones (TPZs) to protect individual trees not in NGZs in accordance with AS4970-2009. Prior to native vegetation removal, confirm that native vegetation and trees to be retained have been adequately protected from impact prior to works. This inspection will be undertaken by an appropriately qualified person (Site Environmental Supervisor) or similar.
Native Vegetation Protection Register	EM03	Maintain a register to identify individual NGZs and TPZs, their specific values, and any design program interactions. NGZs and TPZs will be clearly marked on all plans for the project and be incorporated into the Biodiversity Management Plan
Ongoing Monitoring Of NGZ Fencing	EM04	Conduct regular inspections and maintenance of fencing throughout construction and operation to ensure continued integrity.
Pre-Clearance Survey	EM05	A qualified zoologist must be present to conduct pre-clearance searches in any identified fauna habitats. The pre-clearance protocol is to be developed in consultation with DEECA.
Fauna Salvage	EM06	Prepare a wildlife salvage plan in consultation with appropriate regulator/s. Engage a suitably qualified wildlife handler ('wildlife spotter') holding a relevant and current authorisation under the *Wildlife Act 1975*, to salvage and relocate any wildlife encountered during site clearance works.
Injured Fauna Protocol	EM07	Develop and implement an Injured Wildlife Protocol in consultation with DEECA and wildlife carers which outlines steps to take in the event of injured wildlife being encountered and provides contact details for local wildlife carers.

Measure	Mitigation ID	Action
Hygiene Control	EM08	Appropriate hygiene controls must be implemented throughout the entire project to prevent the spread of environmental and noxious weeds, chytrid fungus and phytophthora. - Ensure that gravel, crushed rock or other non-soil substrate imported to site is free from noxious weed seed - Establish and maintain wash-down locations for machinery, vehicles, tools and footwear. Wash-down locations will: o Be located at least 30 m from waterways and drains. o Be sited close to site entry and exit points. o Collect wastewater and sediment to be disposed of o appropriately. o Include signage to advise personnel of the methods to use. - Drive on surfaced or defined roads and tracks. - Stockpile topsoil (including the top 2 cm of soil along with vegetative material) separately to sub-surface material (sub 2 cm) as the topsoil possesses the greatest risk of weed seed spread from the Project Area. - Retain topsoil within the Project Area to prevent the spread of weeds outside the Project Area. - Assess suitability of cleared vegetation for mulching/erosion protection on a site-by-site basis. Cleared vegetated material containing or with the potential to contain weed seed material will not be used.
Weed Management	EM09	Prepare a Weed Management Plan. Weed controls will include: - Treat high risk weeds prior to works commencing; - Engage experienced land management contractor/s familiar with native plant and weed identification and possess a detailed working knowledge of herbicide selection and use and mechanical removal techniques to undertake weed control. - Implement a monitoring and control program for noxious weed species. - Conduct inspections to check for evidence of weed invasion, degradation or erosion near work areas. - Identify known areas of high threat weed infestations on site plans and in contractor induction material. - Any new infestation of high threat weeds must be controlled as soon as they are detected. - Avoid driving through highly infested areas of weeds, in water or areas of wet soil, where possible - Manage any outbreak of CALP Act and/or Weeds or National Environmental Significance (WoNS) that occurs due to construction activity. Prevent spread into adjacent land. - Ensure induction of all contract staff details the requirements for vehicles and equipment to be free of mud and plant material prior to entering work sites.
Pest Animal Control	EM10	Prepare a pest animal management plan. The pest animal management plan will include: - measures to monitor and control pest fauna (ie. foxes, cats and rabbits) - a requirement for an experienced pest animal contractor who is experienced in working in ecologically sensitive environments to ensure no off-target damage occurs to native fauna (ie. through baiting programs).
Fauna Sensitive	EM11	In areas of potential disturbance to sensitive fauna species the following measures will be implemented:

Measure	Mitigation ID	Action
Lighting Design		- Use of lighting technology that can limit the spread of light outside of the works area and into sensitive habitat (i.e. adjacent woodlands or wetlands) - Using a light beam that is minimally attractive to flying insects. - Facilities established within or close to sensitive habitats will incorporate use of low intensity and longer wavelength lighting such as red/orange (when not conflicting with safety requirements).
Replacement Hollows	EM12	Install nest boxes in adjacent retained habitat to compensate for the removal of hollow-bearing trees. At a minimum, the number of nest boxes installed will exceed the number of hollow-bearing trees removed. Relocation and salvage of suitable habitat (i.e. logs, trunks, stags) that will to be impacted or removed within the Disturbance Footprint will be repurposed and relocated for use as habitat outside of the Disturbance Footprint. Any existing hollows present within trees that are proposed to be impacted within the development footprint should be re-installed within retained vegetation outside of the impact area.
Fauna Escape Mechanisms	EM13	Trenching works will manage any open pits or trenches to reduce potential for fauna entrapment in accordance with the APGA Code of Environmental Practice: Onshore Pipelines (AGPA 2022) through measures such as: - Minimising the length of trench open at a time. - Minimising the amount of time trenches and other excavations are open. - Constructing trench plugs (short section of trench left unexcavated to allow passage of stock or wildlife across the trench) with slopes less than 45 degrees to provide exit ramps for fauna. Provide other exit ramps where practicable and safe to do so. Exit ramps to be installed at the ends of cable trenches. - Creating 'ladders' to enable fauna to exit the excavations (e.g. branches, ropes, planks, floats). - Daily inspections of open trenches by appropriately certified personnel to remove trapped fauna as required. - Checking for fauna prior to backfilling trenches. - Ensuring fauna are discouraged from work areas by erecting barriers where practicable. - Considering installing signage along access routes through conservation reserves to raise awareness of presence of wildlife crossing roads and reduce incidence of wildlife collision. - Developing and implementing a procedure for finding trapped fauna.
Tree And Shrub Removal Timing	EM14	Remove habitat trees or shrubs (particularly hollow-bearing trees) between February and September, where practicable. Habitat trees or shrubs will not be removed between October and January (inclusive) to avoid the breeding season of most nesting birds and mammals unless in accordance with protocols detailed in the Biodiversity Management Plan.
Wildlife-Friendly Fencing	EM15	Establish permanent No Go Zone fencing using post and wire fencing (without barbed wire), or chain mesh fencing 1.8 m high if required. Add a white wire strand at the top of the fence to make it more visible to wildlife. Establish short-term, more temporary NGZs or other fenced areas with a single wire strand with bunting flags rather than orange mesh para-webbing to reduce risk of entanglement.
Traffic Management	EM16	Measures to reduce the risk of wildlife-vehicle interactions will be implemented, including: - Implement speed limits on vehicular traffic along haulage and internal access roads/tracks - Maintain agreed speed limit on access roads to the site as per occupational health and safety requirements.

Measure	Mitigation ID	Action
		- Install signage warning of the presence of wildlife and ensuring signage is maintained. - Record vehicle-wildlife interactions to inform review of management measures. - Installation of canopy bridges over access / haulage roads to facilitate movement of arboreal fauna between habitats.
Seed Collection	EM17	Collect seed from specimens of national and State significant flora to ensure genetic diversity is not lost. Seed to be used to grow specimens for rehabilitation and revegetation works. Any additional seed material to be lodged with the RBGV seedbank.
Progressive Rehabilitation	EM18	Progressively rehabilitate and revegetate areas disturbed by the project with locally indigenous species. Revegetation activities outside the mine and infrastructure footprint should commence as soon as practical to reduce the delay on habitat availability (post-clearance).
Connectivity	EM19	Prioritise the creation of habitat corridor adjacent to the mine footprint to facilitate east/west connectivity to compensate for loss of the existing habitat corridor along Jallumba-Mockinya Road.
Re-Use of Timber	EM20	Retain timber (trunks and logs) from cleared native vegetation for re-use in retained areas and rehabilitated area for use as additional habitat
Re-Use of Rocks	EM21	Retain rocks and relocate to suitable areas to provide habitat for fauna species such as reptiles.
Post-clearance auditing	EM22	Post clearing audit at the end of each stage of development will be undertaken to reconcile the extent of native vegetation impacted during operation and construction against the permitted losses.
Marking Of Access Tracks and Roads	EM23	Access tracks and roads must be clearly marked to prevent the establishment of secondary tracks and indirect native vegetation impacts.
Sedimentation and Erosion Control	EM24	Implement appropriate sediment and erosion control measures prior to any ground-disturbance works and throughout construction and operation to protect retained native vegetation and waterways/waterbodies. Constructed landforms and sedimentation basins must be designed to minimise erosion and sedimentation to off-target areas.
Water Sprays	EM25	Use of water sprays on unsealed roads and surfaces.
Chemical Spills	EM26	Chemicals will be stored and managed in accordance with relevant guidelines and industry best practice
Dust Control	EM27	Develop and implementation of best practice air and dust management controls, including: - Dust suppression sprays and sprinklers - Cessation of works during high-wind conditions

Measure	Mitigation ID	Action
Water Quality	EM28	Water testing must be undertaken bi-annually in Red Gum Swamp and Darragan Swamp in the years after supplementary water has been supplied to ensure no detrimental impacts to water quality and fauna habitats.
Supplementary Watering	EM29	Regulate supplementary flows into Red Gum Swamp and Darragan Swamp to mimic natural variability using an alternate water source. Supplementary water flows to be in accordance with the Supplementary Watering Plan.
Wetland-dependent fauna monitoring	EM30	Annual monitoring of ongoing habitat use by wetland-dependent fauna (frogs, waterbirds, turtles) at Red Gum Swamp and Darragan Swamp
Vegetation Condition Monitoring	EM31	Annual monitoring of vegetation condition and extent at Red Gum Swamp, Darragan Swamp and Jallumba Marsh.
Rehabilitation of Landforms	EM32	Final landform and drainage will be designed and constructed to re-instate pre-disturbance drainage patterns and catchments
Fire Management	EM33	Undertake bushfire risk assessment to identify potential impacts and appropriate mitigation and management measures. Prepare and implement operational controls for fire management, including emergency management responses
Groundwater Monitoring	EM34	Prepare a Groundwater Monitoring Plan including monthly groundwater level monitoring in Project Area to detect extent and risk of changes in groundwater levels and groundwater quality.
Groundwater Management	EM35	The Groundwater Monitoring Plan must include appropriate mitigation and management measures in the event that a risk is identified. Minimise mine dewatering, groundwater drawdown and degradation of groundwater quality
Traffic Noise Management	EM36	Traffic noise levels to not exceed the objectives specified in the VicRoads Traffic Noise Reduction Policy.
High Impact Noise	EM37	High impact noise events limited to daylight hours.
Noise Mitigation	EM38	Where operations or equipment that emit excessive noise or vibrations, control measures such as baffles or mufflers will be employed.
Fauna Sensitivity to Noise	EM39	Prepare a Fauna Management Plan that identifies important areas of habitat for fauna (ie. breeding habitat for Growling Grass Frog) and associated timing in order to mitigate high impact noise and/or vibration activities at key times of year
Placement of stockpiles, machinery and roads	EM40	Place stockpiles, machinery, infrastructure away from waterways, dams and areas supporting native vegetation as much as practical to reduce risk of disturbance.
Biodiversity Management Plan	EM41	Prepare and implement a Biodiversity Management Plan (or equivalent) to outline control measures to protect ecological values and reduce ecological impacts, define the objectives and targets, roles, and responsibilities for management actions. Include measures to

Measure	Mitigation ID	Action
		manage and control impacts on biodiversity from biosecurity threats (weeds, pathogens and pest animals).
Wetland Watering Plan	EM42	Implement the Wetland Watering Plan to ensure no indirect impacts to Jallumba Marsh Flora and Fauna Reserve, Darragan Swamp and Red Gum Swamp as a result of altered surface water hydrology